

CS 506: Introduction to Quantum Computing

Linear Algebra Foundations: Dirac Notation, Operators, and the Spectral Theorem

Yash Dhore

August 31, 2026

1 Overview

This lecture covers more linear algebra used throughout the course, in Dirac notation. The topics covered are as follows: complex amplitudes, the bases used for qubit states, the inner, outer, and tensor products, and linear operators (quantum gates) such as unitary, Hermitian, and projector operators. Then covered is eigenvalues and eigenvectors, the trace and quadratic forms, normal operators, and a preview of the spectral theorem.

2 Complex Numbers and Amplitudes

2.1 Modulus of a Complex Number

A complex number $a + ib$, with $a, b \in \mathbb{R}$, has squared modulus (amplitude)

$$|a + ib|^2 = a^2 + b^2.$$

2.2 Euler's Identity and Normalized Amplitudes

Euler's identity states

$$e^{ix} = \cos x + i \sin x.$$

If $a = \cos x$ and $b = \sin x$ for some real x , then automatically

$$|a + ib|^2 = \cos^2 x + \sin^2 x = 1,$$

and we may write

$$a + ib = \cos x + i \sin x = e^{ix}, \quad |e^{ix}|^2 = 1.$$

Thus, every complex number of unit modulus can be written as e^{ix} for some real x .

3 Quantum State Spaces

3.1 Dimension of a Quantum State

The dimension of an n -qubit quantum state is 2^n for $n = 1, 2, 3, \dots$. This restriction to powers of two is a convention of computation: it is convenient (and finite) to build n -qubit systems out of n two-level qubits. Quantum mechanics itself imposes no such restriction on the dimension of a state space.

Note: every basis used in this course is orthonormal.

3.2 Bases for $n = 1$ Qubit

Name	Dirac Notation	Vector Notation
Standard (computational) basis	$ 0\rangle$	$\begin{pmatrix} 1 \\ 0 \end{pmatrix}$
	$ 1\rangle$	$\begin{pmatrix} 0 \\ 1 \end{pmatrix}$
Hadamard basis	$ +\rangle = \frac{1}{\sqrt{2}} 0\rangle + \frac{1}{\sqrt{2}} 1\rangle$	$\begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$
	$ -\rangle = \frac{1}{\sqrt{2}} 0\rangle - \frac{1}{\sqrt{2}} 1\rangle$	$\begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix}$
General qubit	$\alpha_0 0\rangle + \alpha_1 1\rangle$	$\begin{pmatrix} \alpha_0 \\ \alpha_1 \end{pmatrix}$

Table 1: Bases for one qubit ($n = 1$). For the general qubit, normalization requires $|\alpha_0|^2 + |\alpha_1|^2 = 1$.

3.3 Bases for $n = 2$ Qubits

Name	Dirac Notation	Vector Notation
Standard (computational) basis	$ 00\rangle$	$\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$
	$ 01\rangle$	$\begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$
	$ 10\rangle$	$\begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$
	$ 11\rangle$	$\begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$
Bell basis	$\beta_{00} = \frac{1}{\sqrt{2}} 00\rangle + \frac{1}{\sqrt{2}} 11\rangle$	$\begin{pmatrix} 1/\sqrt{2} \\ 0 \\ 0 \\ 1/\sqrt{2} \end{pmatrix}$
	$\beta_{01} = \frac{1}{\sqrt{2}} 01\rangle + \frac{1}{\sqrt{2}} 10\rangle$	$\begin{pmatrix} 0 \\ 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{pmatrix}$
	$\beta_{10} = \frac{1}{\sqrt{2}} 00\rangle - \frac{1}{\sqrt{2}} 11\rangle$	$\begin{pmatrix} 1/\sqrt{2} \\ 0 \\ 0 \\ -1/\sqrt{2} \end{pmatrix}$
	$\beta_{11} = \frac{1}{\sqrt{2}} 01\rangle - \frac{1}{\sqrt{2}} 10\rangle$	$\begin{pmatrix} 0 \\ 1/\sqrt{2} \\ -1/\sqrt{2} \\ 0 \end{pmatrix}$

Table 2: Bases for two qubits ($n = 2$).

4 Dirac Notation and the Adjoint

4.1 The Adjoint of a Vector

The vector $|\psi\rangle$ is a column vector. We write $\langle\psi|$ for the conjugate transpose (also called the Hermitian conjugate or adjoint) of $|\psi\rangle$:

$$\langle\psi| = |\psi\rangle^\dagger.$$

Example. If

$$|\psi\rangle = \begin{pmatrix} \frac{i}{\sqrt{2}} \\ 1 \\ \frac{1}{\sqrt{2}} \end{pmatrix},$$

then taking the transpose and conjugating each entry gives

$$\langle\psi| = \begin{pmatrix} -\frac{i}{\sqrt{2}} & 1 \\ \frac{1}{\sqrt{2}} \end{pmatrix}.$$

4.2 Inner Product

The dot product is a special case of the inner product. Let

$$|\psi\rangle = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix}, \quad |\Phi\rangle = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{pmatrix}.$$

Then

$$\langle\psi|\Phi\rangle = a_1^*b_1 + a_2^*b_2 + a_3^*b_3 + a_4^*b_4,$$

where a_i^* denotes the complex conjugate of a_i .

4.3 Outer Product

The outer product of $|\psi\rangle$ and $\langle\Phi|$ produces a matrix:

$$|\psi\rangle\langle\Phi| = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix} \begin{pmatrix} b_1 & b_2 & b_3 & b_4 \end{pmatrix} = \begin{pmatrix} a_1b_1 & a_1b_2 & a_1b_3 & a_1b_4 \\ a_2b_1 & a_2b_2 & a_2b_3 & a_2b_4 \\ a_3b_1 & a_3b_2 & a_3b_3 & a_3b_4 \\ a_4b_1 & a_4b_2 & a_4b_3 & a_4b_4 \end{pmatrix}.$$

4.4 Tensor Product

The tensor product of two matrices is used to increase the dimension of a state or operator. For a 2×2 matrix A acting on a 4×4 matrix B ,

$$A \otimes B = \begin{pmatrix} a_1 & a_2 \\ a_3 & a_4 \end{pmatrix} \otimes B = \begin{pmatrix} a_1B & a_2B \\ a_3B & a_4B \end{pmatrix},$$

which is an 8×8 matrix, since each of the four blocks a_iB is itself 4×4 .

5 Linear Operators (Quantum Gates)

5.1 Operators as Matrices

A quantum operator (gate) T acting on n qubits is a $2^n \times 2^n$ square matrix. It must be square for two reasons: the output dimension must equal the input dimension (a constraint of the quantum world), and only square matrices can have inverses. Equivalently, an operator can be viewed as a linear transformation from one vector space to another.

5.2 The Adjoint of an Operator

Example. Let

$$T = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 1 & i \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}.$$

Taking the conjugate transpose,

$$T^\dagger = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 1 & i \end{pmatrix}.$$

Specifically, the first row of T becomes the (conjugated) first column of $T^\dagger$, and the second row of T becomes the (conjugated) second column of $T^\dagger$.

5.3 Unitary Operators

An operator (matrix) U is unitary if

$$U^{-1} = U^\dagger, \quad UU^\dagger = I,$$

where I is the identity matrix. Only unitary operators are allowed in quantum mechanics.

5.4 Example: The NOT Gate

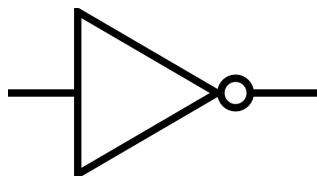


Figure 1: Circuit symbol for the NOT gate.

$$\text{NOT}|0\rangle = |1\rangle, \quad \text{NOT}|1\rangle = |0\rangle,$$

$$\text{NOT} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

5.5 Example: The Identity Gate

$$I|0\rangle = |0\rangle, \quad I|1\rangle = |1\rangle, \quad I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

The identity gate is useful as a no-op: in programming a quantum circuit it is sometimes necessary to do nothing to a qubit on a given time step, and I provides exactly that.

5.6 Hermitian Operators

An operator T is Hermitian (self-adjoint) if

$$T = T^\dagger.$$

5.7 Projector Operators

In this course, an operator P is called a projector if

$$P^2 = P.$$

5.8 Hermitian Projectors Are Unitary

If an operator T is both Hermitian ($T = T^\dagger$) and a projector in the sense of Section 5.7 ($T^2 = T$), then

$$TT^\dagger = T \cdot T = T^2 = T,$$

so T is unitary.

Remark: all eigenvalues of a Hermitian operator are real.

6 Eigenvalues, Eigenvectors, and Operator Decompositions

6.1 Eigenvalue Equation in Dirac Notation

For an operator (matrix) T , a nonzero vector $|x\rangle$ is an eigenvector of T with eigenvalue λ if

$$T|x\rangle = \lambda|x\rangle.$$

We adopt the convention $\langle x|x\rangle = 1$ (i.e. $|x\rangle$ is normalized), so that scalar multiples of an eigenvector (which are also eigenvectors) are not counted as distinct.

6.2 Example: Decomposing the Hadamard Operator

For $n = 1$ (dimension $2^1 = 2$), recall the standard basis $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, and the Hadamard operator

$$H = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}.$$

Splitting H into its four entries as separate matrices,

$$\begin{aligned} H &= \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & \frac{1}{\sqrt{2}} \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ \frac{1}{\sqrt{2}} & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & -\frac{1}{\sqrt{2}} \end{pmatrix}, \\ &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} - \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \end{aligned}$$

and writing each rank-one matrix as an outer product of basis vectors,

$$= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix} + \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \end{pmatrix} + \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix} - \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \end{pmatrix},$$

we arrive at

$$H = \frac{1}{\sqrt{2}}|0\rangle\langle 0| + \frac{1}{\sqrt{2}}|0\rangle\langle 1| + \frac{1}{\sqrt{2}}|1\rangle\langle 0| - \frac{1}{\sqrt{2}}|1\rangle\langle 1|.$$

6.3 Operators in Terms of Basis Vectors

Reminder: any basis can be transformed into any other basis. More generally, for $m = 2^n$, let $|b_1\rangle, |b_2\rangle, \dots, |b_m\rangle$ be any orthonormal basis. Then any operator T can be written as

$$T = \sum_{i=1}^m \sum_{j=1}^m T_{i,j} |b_i\rangle\langle b_j|,$$

where each $T_{i,j}$ is a scalar (in general, complex).

7 Trace of an Operator

7.1 Definition and Example

The trace of a square matrix (operator) A is the sum of its diagonal entries,

$$\text{Tr}(A) = \sum_i A_{ii}.$$

For example, with the Hadamard operator H above,

$$\text{Tr}(H) = \frac{1}{\sqrt{2}} + \left(-\frac{1}{\sqrt{2}}\right) = 0.$$

This also matches

$$\begin{pmatrix} 1 & 0 \end{pmatrix} H \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 & 1 \end{pmatrix} H \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \langle 0|H|0\rangle + \langle 1|H|1\rangle.$$

7.2 Trace in Dirac Notation

How do we write $\text{Tr}(A)$ in Dirac notation? Using an orthonormal basis $|b_1\rangle, \dots, |b_m\rangle$,

$$\text{Tr}(A) = \sum_i \langle b_i | A | b_i \rangle.$$

In particular, using the computational basis for $m = 2^n$,

$$\text{Tr}(A) = \langle 0 | A | 0 \rangle + \langle 1 | A | 1 \rangle.$$

Exercise (left to the reader): show that

$$\langle 0 | A | 0 \rangle + \langle 1 | A | 1 \rangle = \langle + | A | + \rangle + \langle - | A | - \rangle.$$

8 Quadratic Forms

For an $m \times m$ matrix A and vectors $|x\rangle, |y\rangle$,

$$\langle x | A | y \rangle = \begin{pmatrix} x_1 & \cdots & x_i & \cdots & x_m \end{pmatrix} A \begin{pmatrix} y_1 \\ \vdots \\ y_i \\ \vdots \\ y_m \end{pmatrix} = \sum_{i=1}^m \sum_{j=1}^m a_{ij} x_i y_j.$$

This is called a quadratic form. (Side note: when A has entries only on the diagonal and $|x\rangle = |y\rangle$, an expression of this type can be used to define an ellipse.)

9 Normal Operators

An operator A is normal if

$$AA^\dagger = A^\dagger A.$$

9.1 Unitary and Hermitian Operators Are Normal

Unitary operators are normal. If U is unitary, then $UU^\dagger = I$, and

$$U^\dagger U = (UU^\dagger)^\dagger = I^\dagger = I = UU^\dagger,$$

so U is normal.

Hermitian operators are normal. If A is Hermitian, i.e. $A = A^\dagger$, then

$$AA^\dagger = A^2 = A^\dagger A,$$

so A is normal.

10 The Spectral Theorem (Preview)

Let T be a normal operator, with orthonormal eigenvectors $|T_1\rangle, |T_2\rangle, \dots, |T_m\rangle$ and corresponding eigenvalues $T_1, T_2, \dots, T_m$. Then T can be written as

$$T = T_1|T_1\rangle\langle T_1| + \dots + T_m|T_m\rangle\langle T_m| = \sum_{i=1}^m T_i|T_i\rangle\langle T_i|.$$

A full proof, and a worked example, will be given next lecture.

Note: the material up to this point corresponds to what is historically known as Heisenberg's matrix mechanics. Later we will see why this is the same theory as Schrödinger's wave mechanics.

Next class: expand on the spectral theorem and work through an example.